Assignment Beta — Rija Ali — Date:

23K-0057

Q3) a) Probabilities at T = 1: $P_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$

$z_t = [5, 3, 2, 0]$

$z_s = [3, 2.5, 1, 0.5]$

	hospital	clinic	ward	school
P(teacher)	$\frac{e^5}{e^5+e^3+e^2+e^0}$	$\frac{e^3}{e^5+e^3+e^2+e^0}$	$\frac{e^2}{e^5+e^3+e^2+e^0}$	$\frac{e^0}{e^5+e^3+e^2+e^0}$
	↳ 0.8390	↳ 0.1136	↳ 0.0418	↳ 0.0057
P(student)	$\frac{e^3}{e^3+e^{2.5}+e^1+e^{0.5}}$	$\frac{e^{2.5}}{e^3+e^{2.5}+e^1+e^{0.5}}$	$\frac{e^1}{e^3+e^{2.5}+e^1+e^{0.5}}$	$\frac{e^{0.5}}{e^3+e^{2.5}+e^1+e^{0.5}}$
	↳ 0.5483	↳ 0.3325	↳ 0.0748	↳ 0.0450

b) softened probabilities at T = 2: $P_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$

$z_t = [5, 3, 2, 0] \longrightarrow [2.5, 1.5, 1, 0]$

$P_T(\text{hospital}) = \frac{e^{2.5}}{e^{2.5}+e^{1.5}+e^1+e^0} = 0.5977$

$P_T(\text{clinic}) = \frac{e^{1.5}}{e^{2.5}+e^{1.5}+e^1+e^0} = 0.2199$

$P_T(\text{ward}) = \frac{e^1}{e^{2.5}+e^{1.5}+e^1+e^0} = 0.1334$

$P_T(\text{school}) = \frac{e^0}{e^{2.5}+e^{1.5}+e^1+e^0} = 0.0491$

$z_s = [3, 2.5, 1, 0.5] \longrightarrow [1.5, 1.25, 0.5, 0.25]$

$q_T(\text{hospital}) = \frac{e^{1.5}}{e^{1.5}+e^{1.25}+e^{0.5}+e^{0.25}} = 0.4110$

$q_T(\text{clinic}) = \frac{e^{1.25}}{e^{1.5}+e^{1.25}+e^{0.5}+e^{0.25}} = 0.3210$

$q_T(\text{ward}) = \frac{e^{0.5}}{e^{1.5}+e^{1.25}+e^{0.5}+e^{0.25}} = 0.1512$

$q_T(\text{school}) = \frac{e^{0.25}}{e^{1.5}+e^{1.25}+e^{0.5}+e^{0.25}}$

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a) KL Divergence: $\sum p^T \ln(p^T / q^T)$

hospital = $0.5977 \times \ln\left(\frac{0.5977}{0.4110}\right) = 0.2245$

clinic = $0.2199 \times \ln\left(\frac{0.2199}{0.3210}\right) = -0.0825$

ward = $0.1334 \times \ln\left(\frac{0.1334}{0.1512}\right) = -0.0167$

school = $0.0491 \times \ln\left(\frac{0.0491}{0.1177}\right) = -0.0430$

KL = $0.2245 + (-0.0825) + (-0.0167) + (-0.0430)$

= 0.0816

d) $y = [1, 0, 0, 0]$

$CE = -\sum y_i \ln(q_{1,i})$

$CE = -[1\ln(0.5483) + 0\ln(0.3325) + 0\ln(0.0742) + 0\ln(0.0450)]$

= 0.6010

e) Total loss: $L = \alpha T^2 KL(p^T \| q^T) + (1-\alpha) CE(y, q_1)$

$\alpha = 0.5$, $T = 2$, $KL = 0.0816$, $CE = 0.6010$

$L = 0.5 \times 4 \times 0.0816 + 0.5 \times 0.6010 = 0.4637$

f) $\frac{\partial L}{\partial z_i} = \alpha T (q^T - p^T) + (1-\alpha)(q_1 - y)$; $\alpha = 0.5$, $T = 2$

$\nabla L = (q^T - p^T) + 0.5(q_1 - y)$

	hospital	clinic	ward	school
$(q^T - p^T)$	-0.1867	0.1002	0.0178	0.0687
$0.5(q_1 - y)$	-0.2259	0.1663	0.0371	0.0225
∇L	-0.4126	0.2665	0.0549	0.0912

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$\nabla L = [-0.4126, 0.2665, 0.0549, 0.0912]$

g) $z_s' = z_s - 0.5\nabla L$

$= [3, 2.5, 1, 0.5] - 0.5[-0.4126, 0.2665, 0.0549, 0.0912]$

$= [3.2063, 2.3668, 0.9725, 0.4544]$

$KL_{new} = 0.05667$

$CE_{new} = 0.4718$

$L_{new} = 0.5(4)(0.0567) + 0.5(0.4718)$

$= 0.3493$

q4) i) Accuracy (zero-shot) – 95% CI

$= (100 + 80 + 50)/300 = 230/300 = 0.7667$

wald [0.719, 0.815], wilson [0.716, 0.811]

2) Precision, Recall & F1: (zero-shot):

TP = 100, FP = 8 + 20 = 28, FN = 5 + 15 = 20

Positive:

Precision = 100/128 = 0.781

Recall = 100/120 = 0.833

F1 = 2(100)/2(100) + 28 + 20 = 0.806

Negative:

Precision = 80/95 = 0.842

Recall = 80/100 = 0.800

F1 = ~~0.80/2.80~~ = 0.821

Neutral:

Precision = 50/77 = 0.649

Recall = 50/80 = 0.625

F1 = 0.637

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3) Avg. F1:

macro = (0.806 + 0.821 + 0.637) / 3 = 0.755

weighted = (120/300/100) = 0.766

Micro = 0.767 ; for single label multiclass, micro = P = R = F1 = Accuracy

4) Neutral is the weak class, out of 80 true reviews 30 were misclassified 20 as positive & 10 as negative. LLM likely treats mixed-sentiment reviews inconsistently, because neutral was never defined & label name is ambiguous

They should define Neutral explicitly, add few-shot borderline, add a chain-of-thought and collect more borderline annotated data.

5) Stage 2 ;

Accuracy : 106 + 84 + 64 / 300 = 0.8467

↳ wald [0.806, 0.887] , wilson [0.802, 0.883]

Positive:

Precision = 106/121 = 0.876

Recall = 106/120 = 0.883

F1 = 0.880

Negative:

Precision = 84/95 = 0.884

Recall = 84/100 = 0.840

F1 = 0.862

Neutral :

Precision = 64/84 = 0.762

Recall = 64/80 = 0.800

F1 = 0.780

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macro F1 = 0.841 , weighted = 0.847 , micro = 0.847 .

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accuracy rose by 8 points & macro f1 by about 8.6. Neutral class improved most (f1 = 0.637 → f1 = 0.780). Negative recall slipped slightly 0.80 → 0.84 but errors negative → neutral only dropped from 12 → 10.

6, 7, 8 in IPYNB notebook along with all above

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